

WEEK 4 ASSIGNMENT

TITLE:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

OBJECTIVE:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

ASSIGNMENT TASKS

Task 1: Azure Portal and Resource Group

Objective:

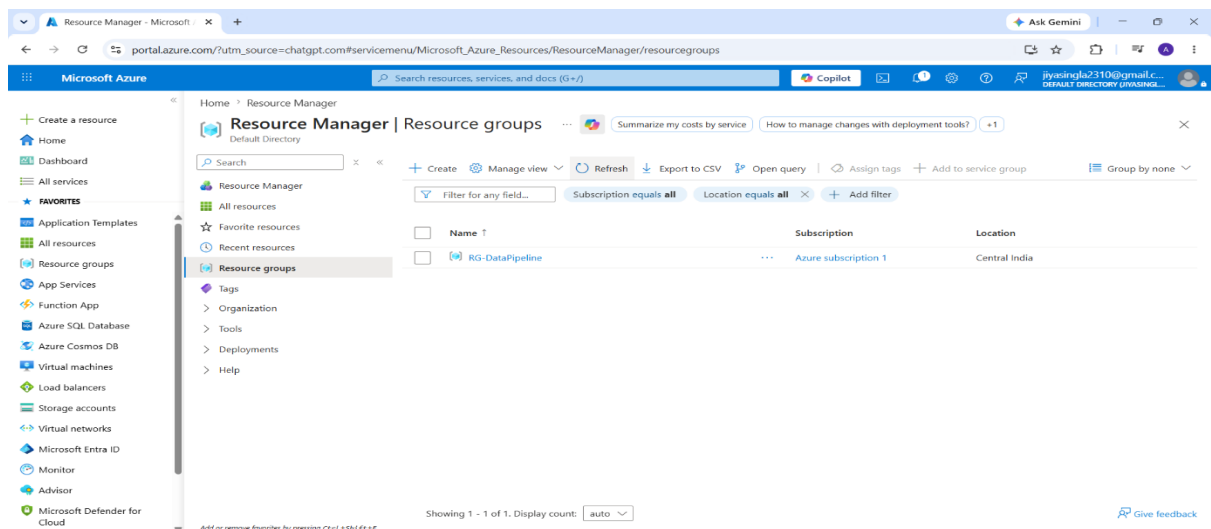
- Explore Azure Portal.
- Create a Resource Group.

Steps Performed

- Logged in to the Microsoft Azure Portal.
- Navigated to the **Resource Groups** service.
- Created a new Resource Group named **RG-DataPipeline**.
- Selected the appropriate subscription and region.

Deliverable:

Screenshot 1.1: Resource Group Overview



Task 2: Storage Setup

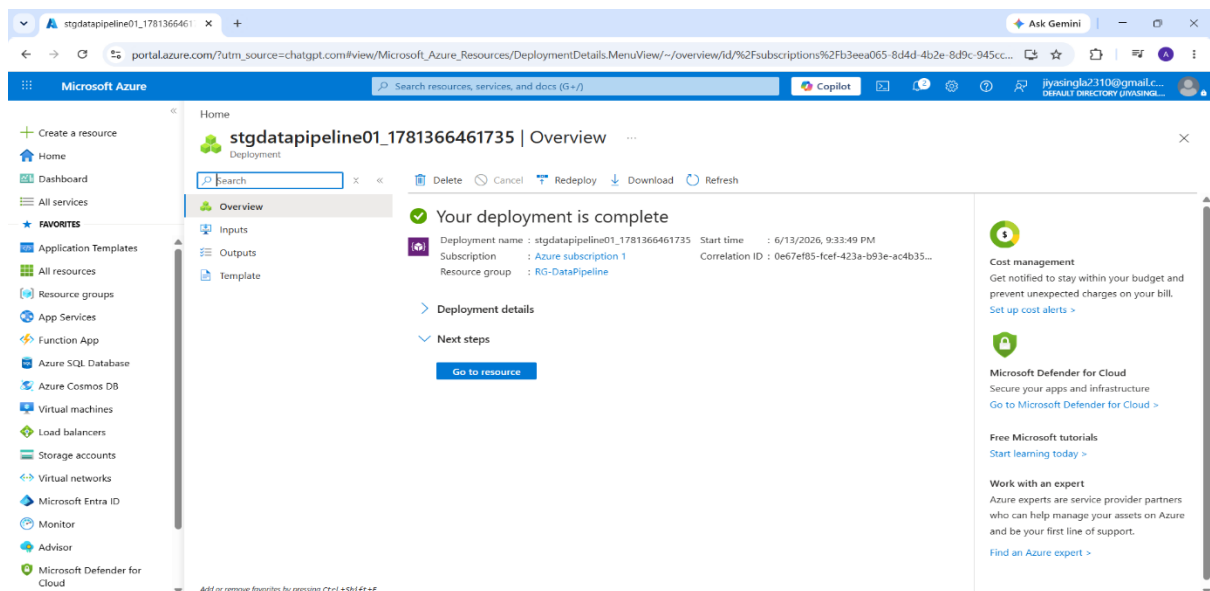
Objective:

- Create a Storage Account.
- Create a Blob Container.
- Upload a CSV file.

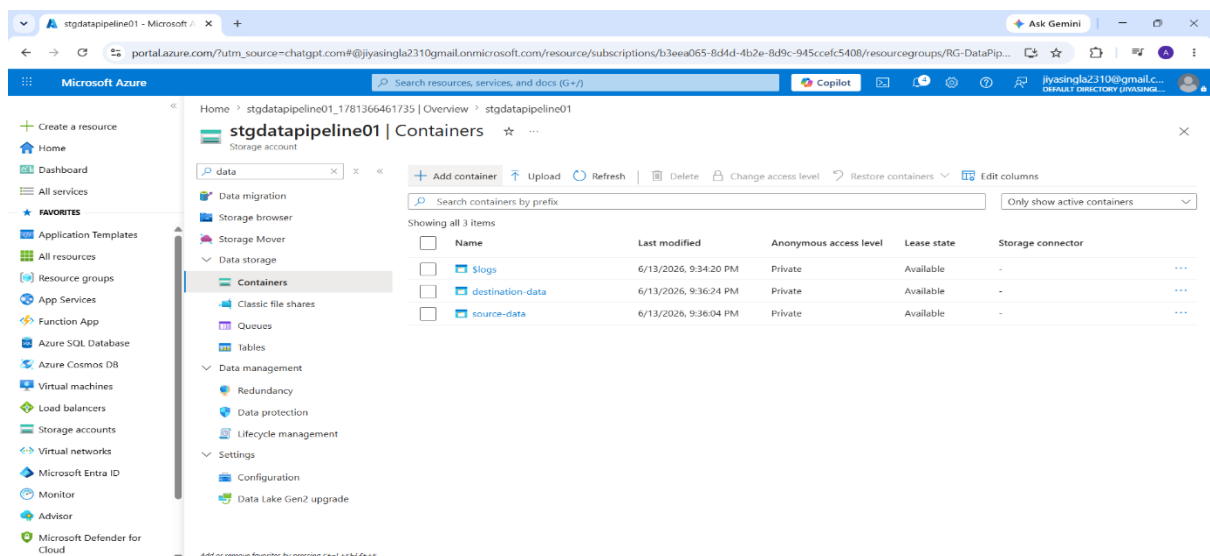
Steps Performed:

- Created an Azure Storage Account.
- Created Blob containers for storing source and destination data.
- Uploaded the **Superstore.csv** file to the source container.

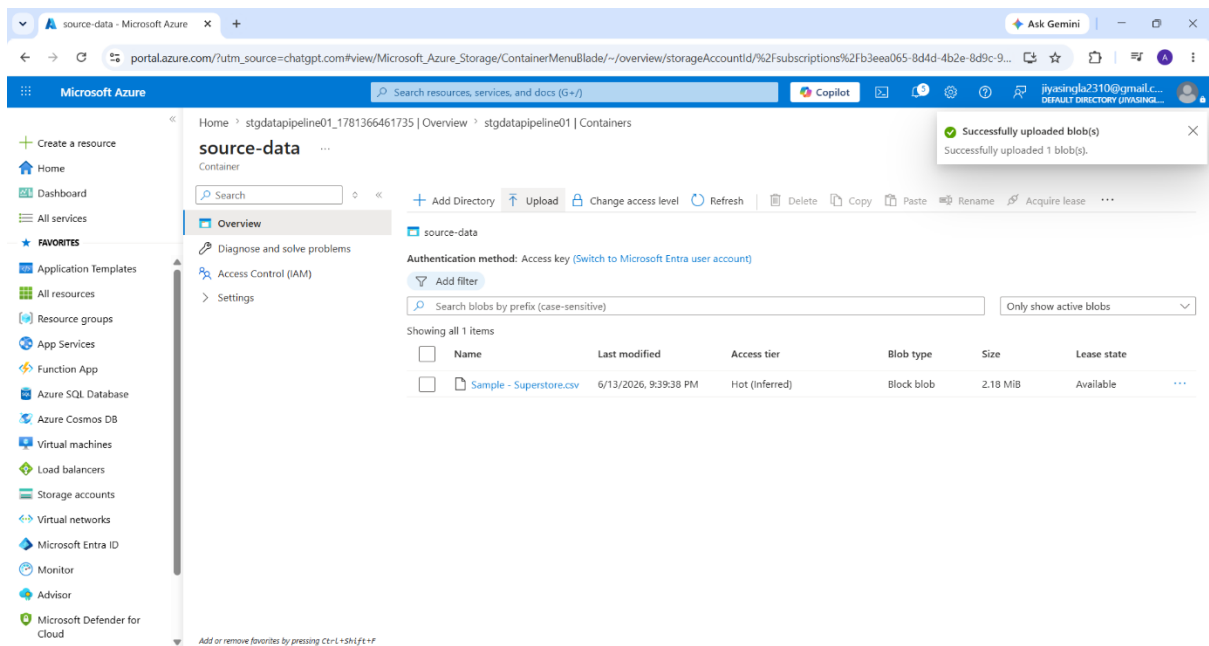
Screenshot 2.1: Storage Account



Screenshot 2.2: Blob Container



Screenshot 2.3: CSV File Uploaded in Blob Container



Task 3: ADF Basics

Objective:

- Create Azure Data Factory.
- Explore ADF UI (Author, Monitor, Manage).
- Create Linked Service (Blob Storage).
- Create source and destination datasets.
- Use Get Metadata activity.

Steps Performed:

- Created an Azure Data Factory instance.
- Explored the ADF Studio interface, including **Author**, **Monitor**, and **Manage** sections.
- Created a Linked Service to connect Azure Blob Storage with ADF.
- Created source and destination datasets for the CSV file.
- Configured the **Get Metadata** activity to retrieve file information.

Deliverables:

Screenshot 3.1A: Azure Data Factory

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation links for Home, Dashboard, All services, Favorites, Application Templates, All resources, Resource groups, App Services, Function App, Azure SQL Database, Azure Cosmos DB, Virtual machines, Load balancers, Storage accounts, Virtual networks, Microsoft Entra ID, Monitor, Advisor, and Microsoft Defender for Cloud. The main content area displays the 'Microsoft.DataFactory-20260613214017 | Overview' page. It shows a deployment status of 'Your deployment is complete' with details: Deployment name: Microsoft.DataFactory.202606132..., Subscription: Azure subscription 1, Resource group: RG-DataPipeline, Start time: 6/13/2026, 9:43:43 PM, and Correlation ID: 39c2029f-34c0-4017-8916-6cafb2... A table lists the resources: 'adf-jiya-datapipeline' of type 'Data factory (V2)' with status 'OK'. The 'Next steps' section includes a 'Go to resource' button. The right sidebar contains informational cards for Cost management, Microsoft Defender for Cloud, Free Microsoft tutorials, and Work with an expert.

Screenshot 3.1B Azure Data Factory Studio Home Page

The screenshot shows the Azure Data Factory Studio home page. The top navigation bar includes links for Plan ahead with Fabric, Azure Data Factory, and a 'Set up code repository' button. The main content area features a large illustration of a data pipeline and four key capabilities: Ingest (Copy data at scale once or on a schedule), Orchestrate (Code-free data pipelines), Transform data (Transform your data using data flows), and Configure SSIS (Manage & run your SSIS packages in the cloud). Below this, a 'Recent resources' section shows 'No items to show'.

Screenshot 3.2 Linked Service - AzureBlobStorage1

The screenshot shows the 'Linked services' page in Azure Data Factory Studio. The left sidebar contains navigation links for General, Connector upgrade advis..., Factory settings, Connections, Linked services, Integration runtimes, Microsoft Purview, ADF in Microsoft Fabric, Source control, Git configuration, ARM template, Author, Triggers, Global parameters, Data flow libraries, Security, Credentials, and Customer managed key. The main content area displays the 'Linked services' section with a 'New' button and a 'Filter by name' input. A table lists the linked services: 'AzureBlobStorage1' of type 'Azure Blob Storage' with 0 related items. A success message 'Successfully created' is displayed in the top right corner.

Screenshot 3.3: Source Dataset

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'source-data' container overview. It includes a search bar, a list of actions (Add Directory, Upload, Change access level, Refresh, Delete, Copy, Paste, Rename, Acquire lease), and a table of blobs. The table has columns for Name, Last modified, Access tier, Blob type, Size, and Lease state. One item is listed: 'Sample - Superstore.csv' with a size of 2.18 MiB and a lease state of 'Available'.

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/13/2026, 9:39:38 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

Screenshot 3.4: Destination Dataset

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'destination-data' container overview. It includes a search bar, a list of actions (Add Directory, Upload, Change access level, Refresh, Delete, Copy, Paste, Rename, Acquire lease), and a table of blobs. The table has columns for Name, Last modified, Access tier, Blob type, Size, and Lease state. One item is listed: 'Sample - Superstore.csv' with a size of 2.18 MiB and a lease state of 'Available'.

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/13/2026, 10:32:08 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

Screenshot 3.5 Pipeline-Get Metadata Activity

The screenshot shows the Microsoft Azure Data Factory pipeline editor. The left sidebar contains navigation options like 'Factory Resources', 'Activities', and 'Copy data'. The main content area displays the pipeline design canvas. It includes a 'Get Metadata' activity (Get Metadata1) and a 'Copy data' activity (Copy data1). The 'Copy data' activity is configured with 'Copy data1' as the sink. The 'Mapping' tab is selected, showing 'Type conversion settings' and 'Import schemas' options. A 'Factory validation output' dialog box is open, indicating 'Publishing completed' and 'Successfully published'.

Task 4: Pipeline Development

Objective:

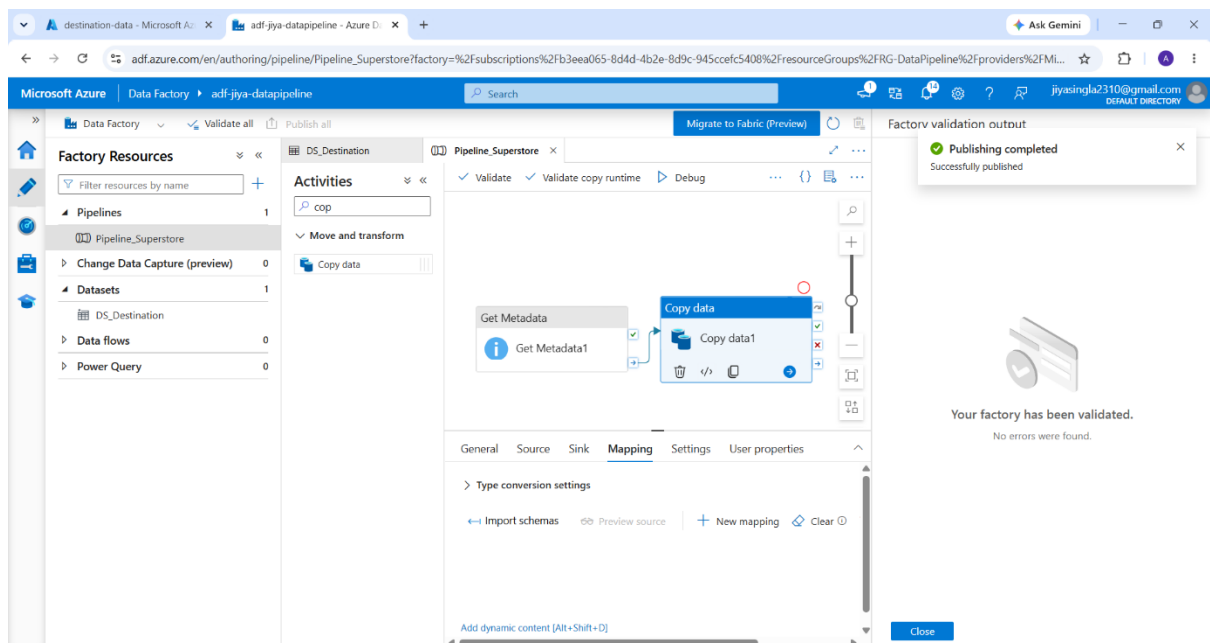
- Create a pipeline using the Copy Data activity.
- Configure source and destination datasets.
- Add a ForEach activity (optional for multiple files).

Steps Performed:

- Created a pipeline in Azure Data Factory.
- Added the **Get Metadata** activity to validate file information.
- Added the **Copy Data** activity to copy the CSV file from the source container to the destination container.
- Configured the source and destination datasets.
- Connected the activities to create an end-to-end data pipeline.

Deliverable:

Screenshot 4.1: Pipeline Design



Task 5: Pipeline Execution

Objective:

- Execute the pipeline using Debug/Trigger.

Steps Performed:

- Executed the pipeline using the **Debug** option in Azure Data Factory.
- Monitored the pipeline execution in the **Monitor** section.
- Verified that the pipeline execution completed successfully with the status **Succeeded**.

Deliverable:

Screenshot 5.1: Pipeline Execution Status (Succeeded)

The screenshot shows the 'Pipeline runs' section in Azure Data Factory. It displays a table with two rows, both showing a 'Succeeded' status. The table includes columns for Pipeline name, Run start, Run end, Duration, Status, Triggered by, Run ID, and Parameters.

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
Pipeline_Superstore	6/13/2026, 10:59:33	6/13/2026, 11:00:05	33s	Succeeded	Manual trigger	4d8bc439-ed9e-43ce...	
Pipeline_Superstore	6/13/2026, 10:58:15	6/13/2026, 10:58:53	38s	Succeeded	Manual trigger	75c41c2e-be46-4213...	

Screenshot 5.2: Activity Run Details

The screenshot shows the 'Activity runs' section in Azure Data Factory. It displays a table with two rows, both showing a 'Succeeded' status. The table includes columns for Activity name, Activity status, Activity name, Run start, Duration, Integration runtime, User properties, and Activity run ID.

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime	User properties	Activity run ID
Copy data1	Succeeded	Copy data	6/13/2026, 10:58:36 PM	16s	AutoResolveIntegrationRuntime (Central India)		8a6b80c0-b785-49c...
Get Metadata1	Succeeded	Get Metadata	6/13/2026, 10:58:22 PM	13s	AutoResolveIntegrationRuntime (Central India)		40826cd0-e0ef-4b52

Task 6: IAM Roles

Objective:

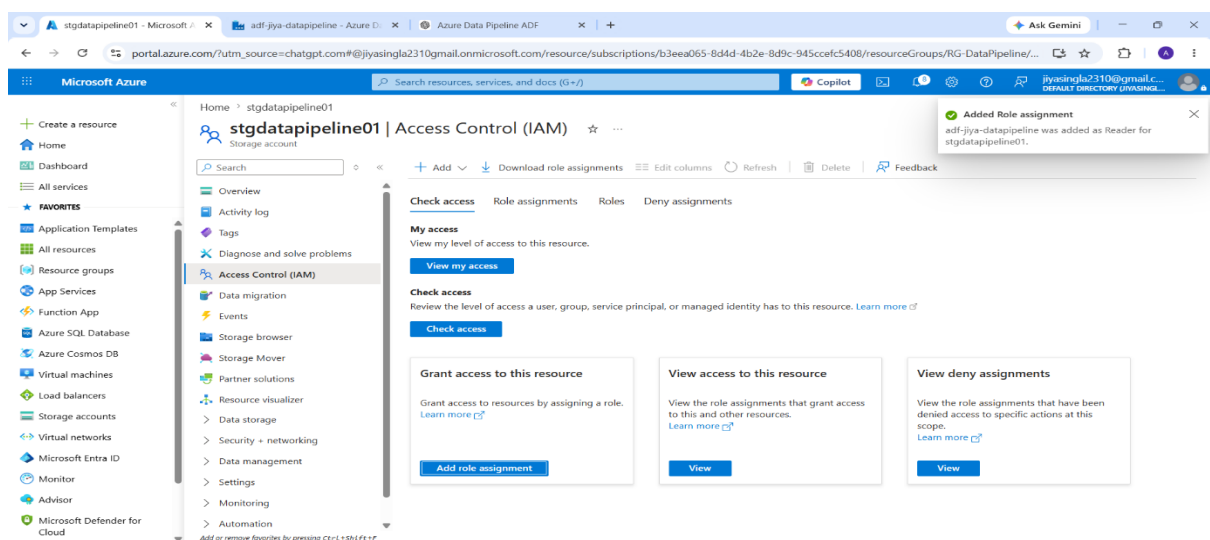
- Assign Reader and Contributor roles.
- Provide Azure Data Factory access to the Storage Account.

Steps Performed:

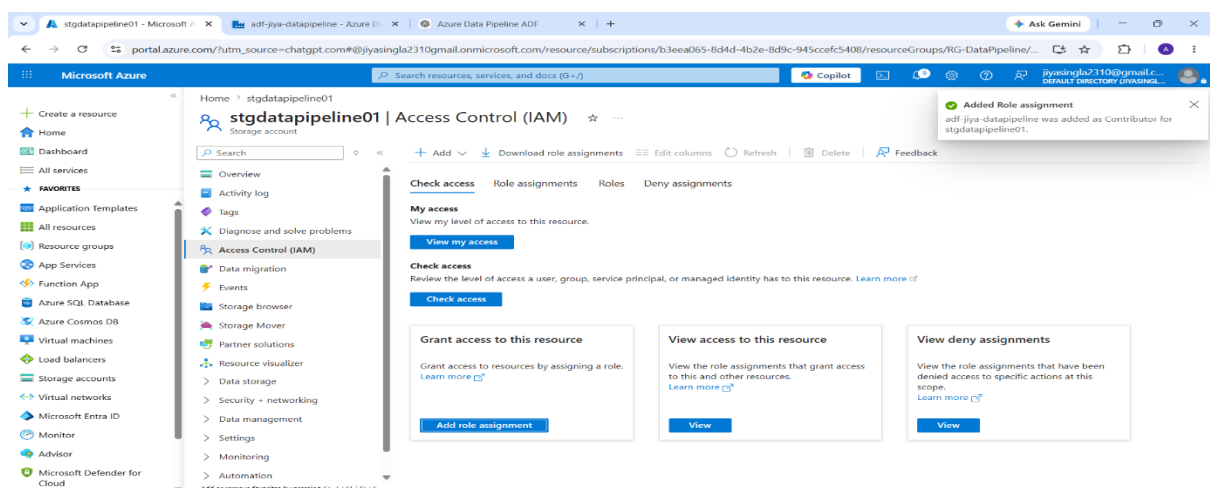
- Navigated to the **Access Control (IAM)** section of the Storage Account.
- Assigned the **Reader** role to enable read access.
- Assigned the **Contributor** role to enable management access.
- Granted Azure Data Factory access to interact with Blob Storage.

Deliverables:

Screenshot 6.1 IAM Reader Role Assignment



Screenshot 6.2: IAM Contributor Role Assignment



Mini Project (Final Task)

Problem Statement

Build a complete pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory.

Requirements

- **Source:** CSV file stored in Azure Blob Storage.
- **Components Used:** Linked Service, Dataset, and Pipeline.
- **Process:** Get Metadata activity for file validation and Copy Data activity for data transfer.
- **Destination:** New file/location in Blob Storage.

Implementation

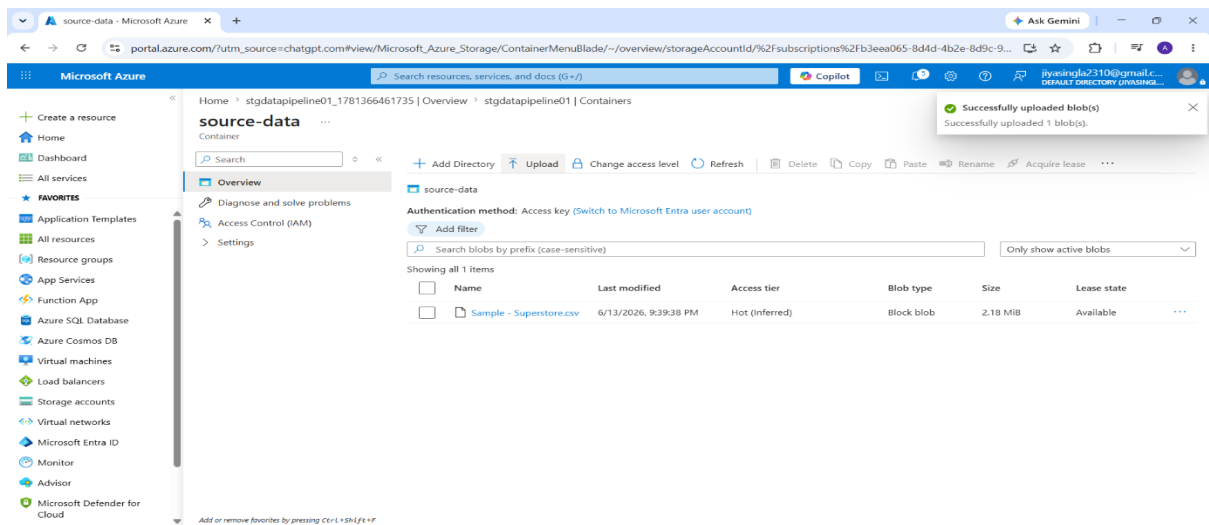
- Uploaded the **Superstore.csv** file to the source Blob container.
- Connected Azure Blob Storage to Azure Data Factory using a Linked Service.
- Created source and destination datasets.
- Used the **Get Metadata** activity to validate file information.
- Used the **Copy Data** activity to copy the file to the destination container.
- Executed and monitored the pipeline successfully.

Expected Output

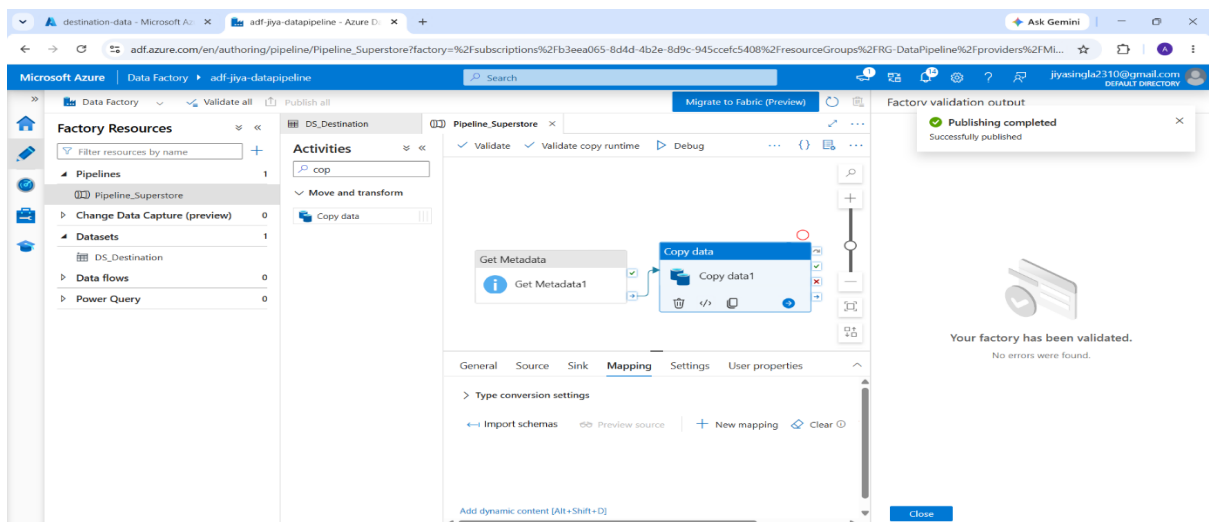
- Pipeline executed successfully.
- Data copied to the destination container.
- Metadata validated successfully.

Deliverable

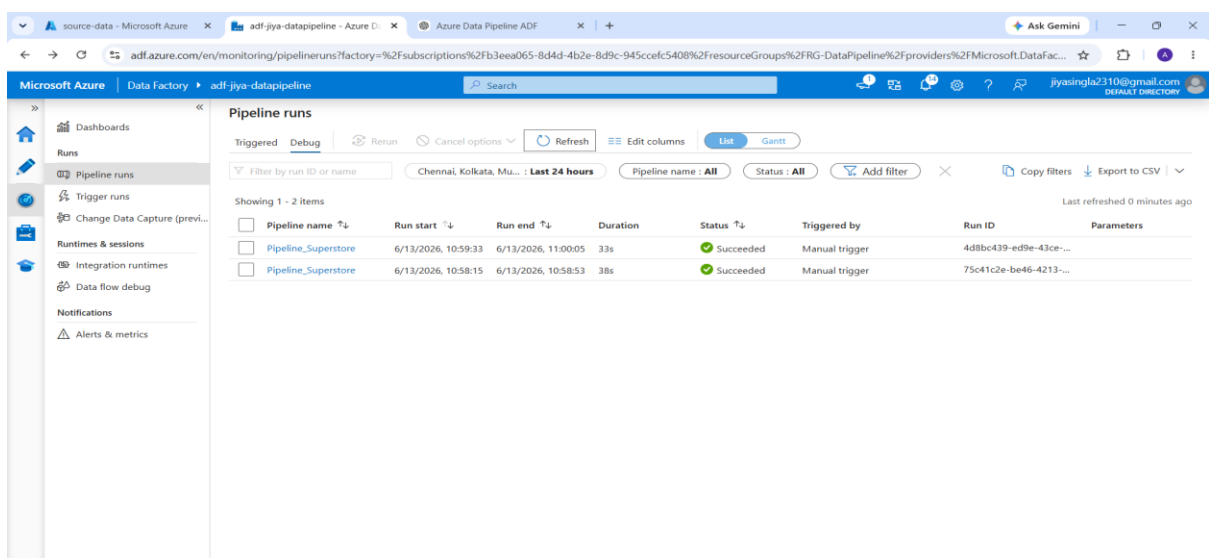
Screenshot MP.1: Source CSV File in Blob Storage



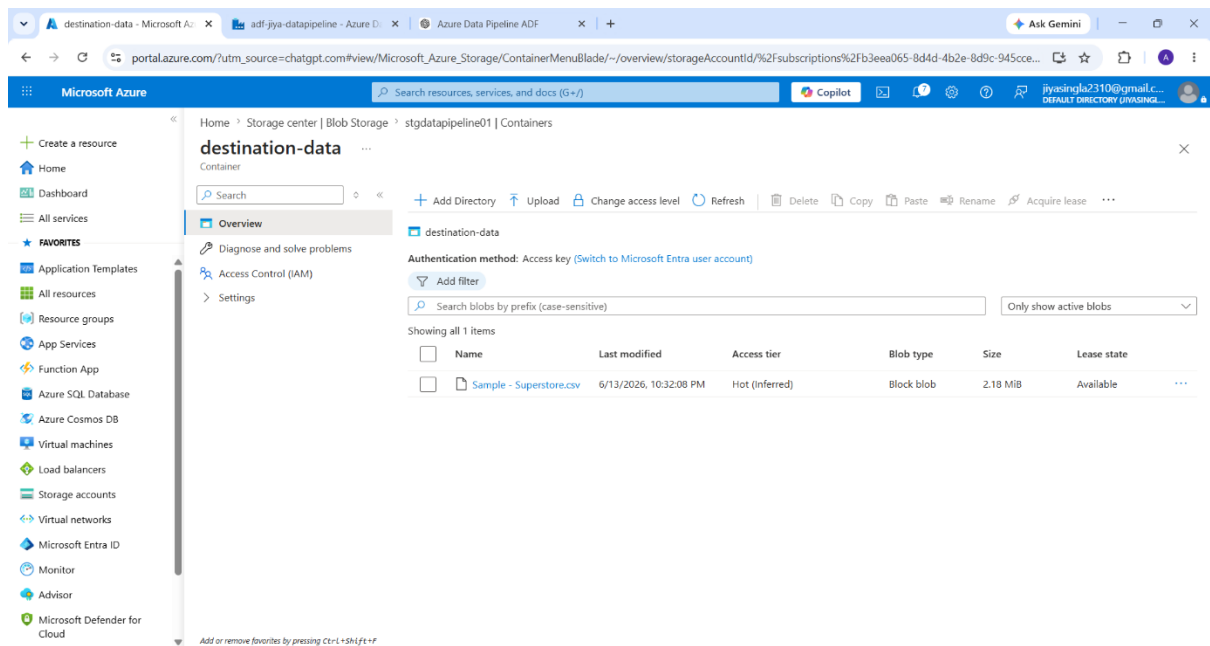
Screenshot MP.2: Pipeline Design with Metadata Validation



Screenshot MP.3: Pipeline Execution Succeeded



Screenshot MP.4: Destination File/Container



BRIEF SUMMARY:

This project demonstrated the implementation of an end-to-end data pipeline using Azure Blob Storage and Azure Data Factory (ADF). A Resource Group, Storage Account, and Blob Containers were created in Azure to manage and store data efficiently. The Superstore CSV dataset was uploaded to Blob Storage and integrated with ADF through Linked Services and Datasets.

The Get Metadata activity was configured to validate file properties such as existence and file information, while the Copy Data activity was used to transfer data from the source container to the destination container. The pipeline was executed successfully using the Debug/Trigger option, and its execution was monitored in ADF.

Additionally, IAM roles such as Reader and Contributor were assigned to ensure secure access between Azure Data Factory and Storage resources. The successful execution of the pipeline confirmed that the data was copied correctly and metadata validation was completed. This project provided hands-on experience with Azure cloud services, data integration, pipeline orchestration, and access management using Azure Data Factory.